Banking Portfolio Risk Analytics Report

1. Problem Statement

Develop a basic understanding of risk analytics in banking and financial services and understand how data is used to minimise the risk of losing money while lending to customers. Financial institutions face ongoing challenges in balancing loan portfolio growth with exposure to default risk. By establishing data-driven analytics frameworks, banks can effectively evaluate applicant profiles, monitor portfolio distribution, and optimize credit issuance parameters.

2. Solution

With our dashboards which are created using Power BI latest tools helps the company to make a decision based on the applicant's profile like if the applicant is likely to repay the loan then approving the loan otherwise not.

The interactive analytics platform aggregates multi-table relational data to provide decision-makers with comprehensive visibility into customer profiles, total liabilities, asset distributions, and revenue generation metrics across global regions.

3. About Dataset

This dataset basically contains information about bank details, various client details which consists of multiple tables which are interlinked with each other through keys like primary key and foreign key.

The relational structure comprises five core tables:

- **Banking Relationship:** Tracks account associations and institution classifications.
- **Client-Banking:** Contains core applicant demographics, financial balances, and loan figures.
- **Gender:** Reference table for demographic segmentation.
- **Investment Advisor:** Maps clients to assigned wealth and advisory professionals.
- **Period:** Temporal lookup table for historical trend analysis.

4. Data Cleaning

To prepare the dataset for reliable reporting and model ingestion, the following transformation steps were executed:

- **Engagement Timeframe:** Creating a new column Engagement Timeframe in client-banking column which tells about the time line of the clients in banks.
- **Engagement Days:** Creating a new column Engagment Days in Client-Banking table how many days the client spent from the date of joining in banks.
- **Income Band Binning:** Creating bins for the Estimated Income < 100000 as low and <300000 as Mid with the column named as Income Band in Clients-Banking table.
- **Processing Fees Logic:** Creating a new column named as Processing Fees for the column Fee Structure like if fee structure is high then processing fee would be 0.05.

5. Calculated Functions & KPI's

The business logic and metrics are constructed using Data Analysis Expressions (DAX). Below is the comprehensive catalog of key performance indicators and calculated measures:

Metric Name	DAX Expression	Description
Total Clients	`Total Clients = DISTINCTCOUNT('Clients - Banking'[Client ID])`	Unique count of registered banking clients.
Total Loan	`Total Loan = [Bank Loan] + [Business Lending] + [Credit Cards Balance]`	Combined total exposure across all credit products.
Bank Loan	`Bank Loan = SUM('Clients - Banking'[Bank Loans])`	Aggregate sum of standard term bank loans.
Business Lending	`Business Lending = SUM('Clients - Banking'[Business Lending])`	Aggregate value of commercial and business loans.

Metric Name	DAX Expression	Description
Total Deposit	`Total Deposit = [Bank Deposit] + [Savings Account] + [Foreign Currency Account] + [Checking Accounts]`	Total client funds held across all account types.
Total Fees	`Total Fees = SUMX('Clients - Banking' , [Total Loan] * 'Clients - Banking'[Processing Fees])`	Estimated revenue generated from processing fees.
Bank Deposit	`Bank Deposit = SUM('Clients - Banking'[Bank Deposits])`	Standard bank deposit holdings.
Checking Account Amount	`Checking Accounts = SUM('Clients - Banking'[Checking Accounts])`	Total balance held in transactional checking accounts.
Total CC Amount	`Total CC Amount = SUM('Clients - Banking'[Amount of Credit Cards])`	Total credit limit allocated across credit cards.
Saving Account Amount	`Savings Account = SUM('Clients - Banking'[Saving Accounts])`	Accumulation in dedicated savings accounts.
Foreign Currency Amount	`Foreign Currency Account = SUM('Clients - Banking'[Foreign Currency Account])`	Total holdings converted from international currencies.

Metric Name	DAX Expression	Description
Engagement Account	`Engagment Length = SUM('Clients - Banking'[Engagment Days])`	Cumulative client engagement duration in days.
Credit Cards Balance	`Credit Cards Balance = SUM('Clients - Banking'[Credit Card Balance])`	Current outstanding debt on credit card accounts.

6. Visualization And Result

The reporting suite consists of four specialized dashboard views designed for executive and operational analysis:

Dashboard Structure

- **Home:** Executive overview displaying high-level portfolio KPIs and quick navigation.
- **Loan Analysis:** In-depth breakdown of credit exposure, risk distributions, and default drivers.
- **Deposit Analysis:** Overview of liquidity, account types, and capital retention across segments.
- **Summary Dashboard:** Consolidated cross-functional view comparing overall lending vs. deposit performance.

Key Observations

- **Income Band Distribution:** Bank loan is highest for Mid Income band and lowest for High Income band.
- **Geographic Feature Density:** Highest features are in the European countries followed by Asian Countries; Lowest features are in Australian countries followed by African countries.
- **Regional Exposure:** Bank loan is highest for European countries and lowest for Australian countries.

7. Conclusion & Future Work

Conclusion

Empowered by the latest data visualization techniques, Power BI dashboards are among the most effective resources for using in banking sector. As outlined in this write-up, a banking operations dashboard in Power BI can be developed with key banking related metrics and KPIs.

Future Work

With these dashboards banks can easily know what is the total loan amount and all other things of a particular investor. It also helps which type of banks have more number of clients as we can see private banks have more number of clients so it can helps other banks can build their strategies to increase clients. It also provides insights about which nationality has highest bank loans. It gives information about various types of amount involved in different types of accounts by investors.